MATH3802
201767438

Ananth Eswaran

November 2025

Abstract

This report examines monthly water levels in a Hampshire pond with the aim of understanding how they change over time and to build an explanatory model for these changes. The overall level is fairly stable, apart from a noticeable dip during the late 2000s that appears to be a temporary event. To focus on genuine month-to-month fluctuations, I first remove small seasonal patterns and any long-term effects that would otherwise hide the underlying behaviour. This allows us to identify how each month's water level relates to the previous months without the influence of seasonal trends. After comparing various possible models, the best model is most likely the one that uses the past two months' level. This final model gives a clear picture of the pond's underlying behaviour and can help in understanding or forecasting short-term changes.

1 Introduction

1.1 Authors Note

The following research follows a dialectic methodology. Sub-sections can be thought of as following a linear thinking style: context → thesis → testing → inferred conclusion. Keeping this in mind greatly improves the readability of the literature. However, due to space restrictions, the research has the drawback of limited testing methods. Ideally, a high-level inference of the best model requires a far more rigorous approach. With that context, the conclusions presented in the following paragraphs can be viewed as recommendations.

1.2 Background

To begin, we consider a monthly series of pond water levels. Pictorially examining the data provides immediate rough ideas of its nature. At first glance, the series looks roughly stationary around its mean; there is no strong upward or downward trend visible over the decades shown. However, there is a noticeable dip around the late 2000s, but considering it does not appear to be followed by a lasting drop in level, it is likely a shock rather than a structural change. The variance seems to be mostly stable (homoscedastic), and there is evidence of potential seasonality with peaks occurring approximately once a year.

1.3 Initial Thesis

Given the context, it is likely that the series contains little or no long-term trend but may exhibit seasonal patterns or short-term persistence. A common starting point in time series analysis is to decompose the observations X_t into trend, seasonal, and irregular components. The general additive decomposition can be written as

$$X_t = \mu(t) + s(t) + \varepsilon_t, \tag{i}$$

where $\mu(t)$ denotes the long-term trend, $s(t)$ represents a seasonal effect, and ε_t captures short-term fluctuations or noise.

The seasonal component is assumed to exhibit strict periodicity, i.e.

$$s(t) = s(t + a), \quad \forall t, \tag{ii}$$

where a is the known seasonal period (for example, $a = 12$ for monthly data with an annual cycle).

In situations where the empirical behaviour of the series suggests little or no long-term trend, the trend component $\mu(t)$ may be treated as approximately constant. Under this assumption, the decomposition simplifies to

$$X_t = s(t) + \varepsilon_t, \tag{iii}$$

since a constant trend can be absorbed into the mean level of the seasonal pattern. In such cases, any systematic structure in the series is primarily attributable to seasonal variation, while the remaining variability reflects short-term persistence or random fluctuations.

2 Data Inspection and model suggestions

2.1 Regression and trend identification

In order to test the above thesis, we run a linear regression on the time series. Figure 2(a) reveals an almost flat slope, potentially indicative of no long-term linear trend. The estimated regression function is

$$u(t) = \hat{\beta}_0 + \hat{\beta}_1 X_t,$$

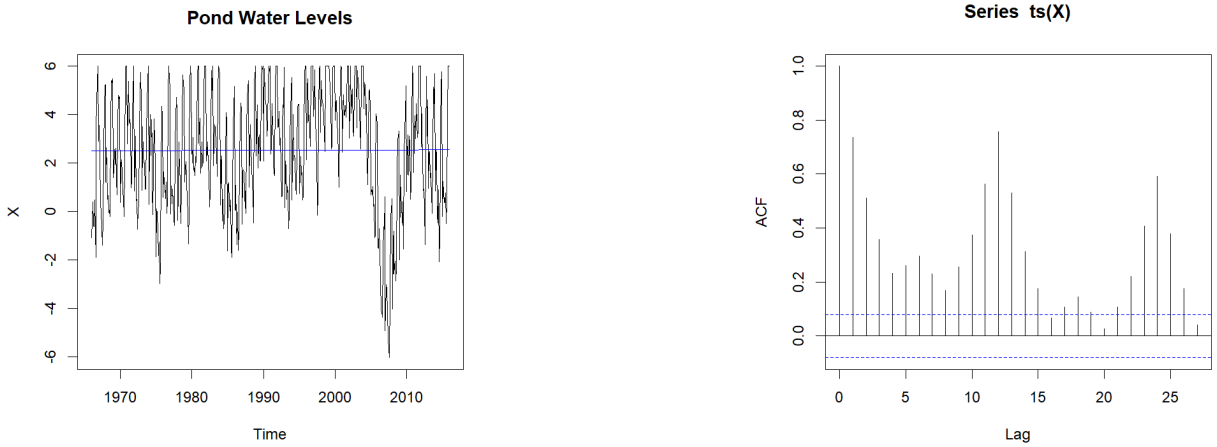
where

$$\hat{\beta}_0 = 2.498 \qquad (p < 2 \times 10^{-16}),$$
$$\hat{\beta}_1 = 6.657 \times 10^{-5} \qquad (p = 0.909).$$

Explicitly, the fitted model is

$$u(t) = 2.498 + 6.657 \times 10^{-5}\, X_t.$$

Our p-values for our model reveal a highly significant intercept at a level of significance of 0.1% for a t-test. In contrast, the slope is statistically insignificant at p=0.909, which is even above the level of significance of 10%. An insignificant slope refutes the idea of there being a long-term trend, for if there were, we would expect a significant (changing) slope, which reflects something moving in time. $u(t)$ here does indeed decompose to a constant; in line with (iii). It would also be useful to note that we do not consider a log regression model due to the variance appearing (Figure 1(a)) generally stable.



(a) Trend and regression *(b) ACF of pond levels*

Figure 1: Trend and autocorrelation diagnostics

To test our presumption of seasonality, it would help to plot and interpret the autocorrelation function (ACF) of the time series X. The ACF (Figure 1) exhibits a notably high first-lag value, indicating that each observation is strongly dependent on its immediate predecessor. Furthermore, the presence of periodic peaks approximately every 12 lags (excluding lag 0) suggests the possibility of annual seasonality. However, definitive conclusions require formal testing of the statistical significance of the relevant model parameters.

2.2 Seasonal effects modeling

The seasonal component repeats every n periods and can be represented as

$$s(t) = s_0,\, s_1,\, \ldots,\, s_{n-1} \quad \text{repeating.}$$

A common way to estimate these seasonal effects is through the use of dummy variables. In our formulation, the expanded model of (i) can be written as

$$X_t = \mu(t) + \sum_{i=0}^{11} s_i\, \delta_{t,i} + \varepsilon_t, \tag{iv}$$

which, under (iii)

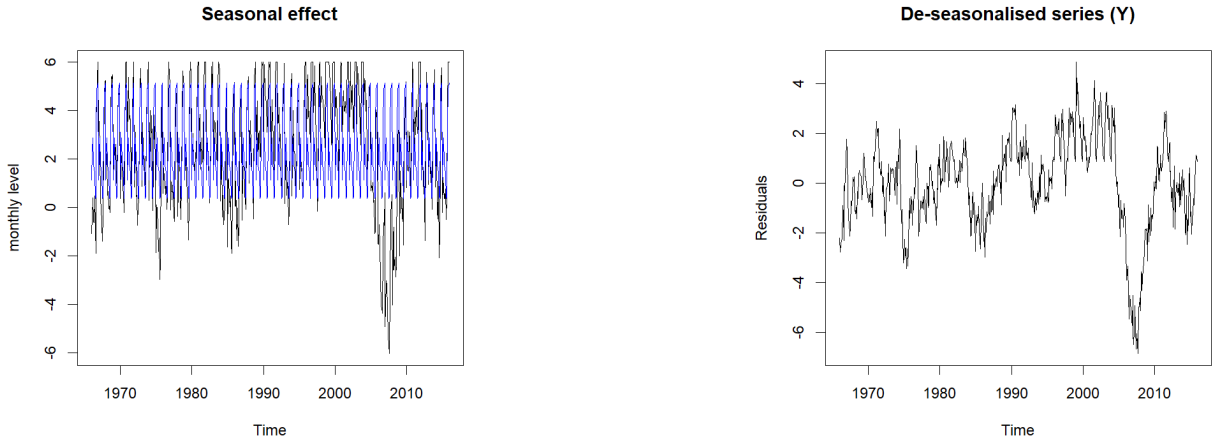
$$X_t = \sum_{i=0}^{11} s_i\, \delta_{t,i} + \varepsilon_t, \tag{v}$$

Month	Jan (s_0)	Feb (s_1)	Mar (s_2)	Apr (s_3)	May (s_4)	Jun (s_5)
p-value	2.27×10^{-5}	$< 2 \times 10^{-16}$	$< 2 \times 10^{-16}$	1.10×10^{-10}	1.52×10^{-9}	1.93×10^{-7}
Significance (%)	0.0023	< 0.000001	< 0.000001	0.000011	0.000152	0.0193

Month	Jul (s_6)	Aug (s_7)	Sep (s_8)	Oct (s_9)	Nov (s_{10})	Dec (s_{11})
p-value	0.166	0.120	$< 2 \times 10^{-16}$	$< 2 \times 10^{-16}$	$< 2 \times 10^{-16}$	$< 2 \times 10^{-16}$
Significance (%)	16.6	12.0	< 0.000001	< 0.000001	< 0.000001	< 0.000001

Table 1: P-values and significance levels for seasonal dummy coefficients.

where $s_0, s_1, \ldots, s_{11}$ represent the twelve monthly seasonal effects, and $\delta_{t,i}$ is a dummy variable taking the value 1 when observation t falls in month i, and 0 otherwise. Performing a t-test on our dummy variables reveals 10 out of 12 coefficients are significant at the 0.1% level. Our seasonal effect model (Figure 2(a)) is therefore a highly accurate estimator of seasonality. As per (v) our $u(t)$ has been reduced and it follows that the remainder after subtracting the seasonal effect are the series of the residuals ε_t. Let us denote these by Y (Figure 2(b)).



(a) Seasonal effect

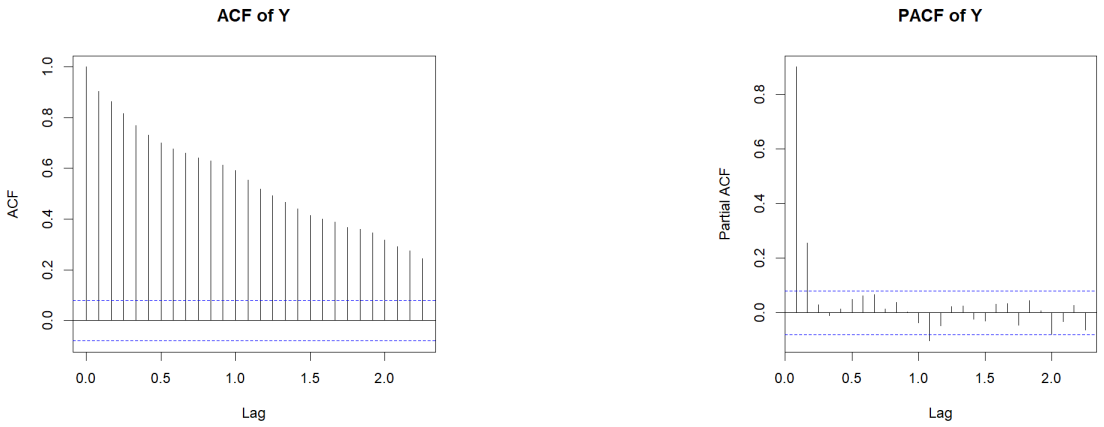
(b) De-seasonalised series

Figure 2: Seasonal and de-seasonalised dtructure

2.3 Residuals model recommendation

The residuals of the time series contain its true stochastic mechanism; observe how the de-seasonalised series (Figure 2(b)) are the residuals. Capturing that mechanism is essential for accurate inference and reliable

forecasting. The Autocorrelation function shows the correlation between a time series and its lagged values, revealing structural patterns.



(a) ACF *(b) PACF*

Figure 3: Autocorrelation and Partial Autocorrelation of residuals

The autocorrelation function (ACF) of a time series measures linear dependence between observations separated by k time steps. Different model classes produce characteristic ACF shapes, which help distinguish autoregressive (AR) from moving-average (MA) behaviour.

The ACF of Y has a very high lag 1 value, then decays gradually and smoothly along subsequent lags. The PACF has significant early spikes, highlighting that the observations depend on their immediate previous values. This behaviour is characteristic of an auto-regressive $AR(p)$ structure-often $AR(1)$ or $AR(2)$-because AR models typically exhibit a slowly declining, roughly geometric decay in the ACF. In contrast, a moving-average $MA(q)$ process would display a few significant auto-correlations followed by an abrupt cutoff after lag q, which is clearly not consistent with the pattern observed in Figure 3.

3 AR(p) testing

3.1 Solving for coefficient estimates

Solving the Yule–Walker equations yields the following parameter estimates:

$$\text{AR(1): } \hat{\phi}_1 = 0.9025, \qquad \hat{\sigma}^2 = 0.6294, \quad Z_t = 0.9025\, Z_{t-1} + \varepsilon_t,$$

$$\text{AR(2): } (\hat{\phi}_1, \hat{\phi}_2) = (0.6709,\, 0.2566), \qquad \hat{\sigma}^2 = 0.589, \quad Z_t = 0.6709\, Z_{t-1} + 0.2566\, Z_{t-2} + \varepsilon_t,$$

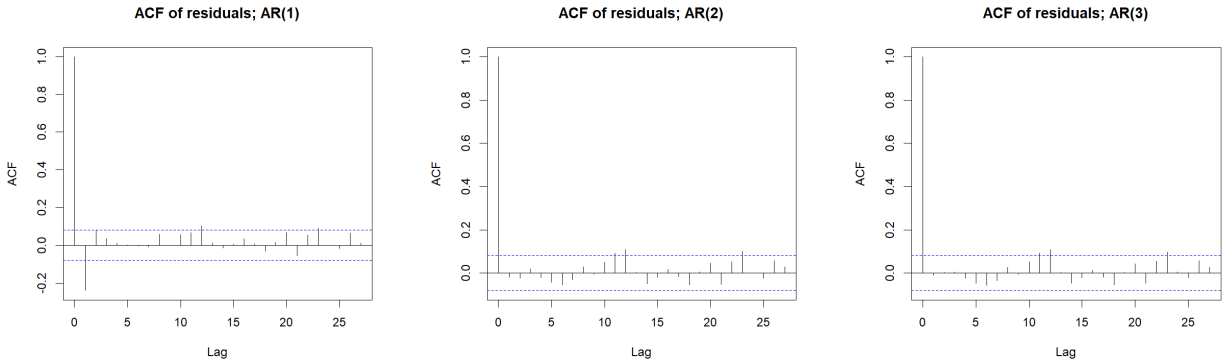
$$\text{AR(3): } (\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3) = (0.6635,\, 0.2373,\, 0.0287), \qquad \hat{\sigma}^2 = 0.5895, \quad Z_t = 0.6635\, Z_{t-1} + 0.2373\, Z_{t-2} + 0.0287\, Z_{t-3} + \varepsilon_t.$$

3.2 Determining optimal model

The ideal AR(p) order is the one that produces residuals that are indistinguishable from white noise , i.e., residuals that are uncorrelated, have a zero mean, and have a constant variance.

Pictorial interpretation

Note that often the graphs produced by R can be misleading. The "blue lines" in Figure 4 are an approximation of the 95% confidence interval, meaning that it can be visually incorrect sometimes. Even then, it is very clear when observing the AR(1) residuals' ACF plot, that there is a significant correlation at the first lag. Contrarily, the plots corresponding to AR(2) and Ar(3) have most of their lag spikes well within the 95% confidence interval. We can safely exclude the AR(1) model from our consideration given the glaring -0.25 lag 1 spike. The consideration, then, is between the AR(2) and AR(3) models.



(a) AR(1) residual ACF *(b) AR(2) residual ACF* *(c) AR(3) residual ACF*

Figure 4: Autocorrelation functions of AR(p) residuals for $p = 1, 2, 3$.

White noise and significance testing

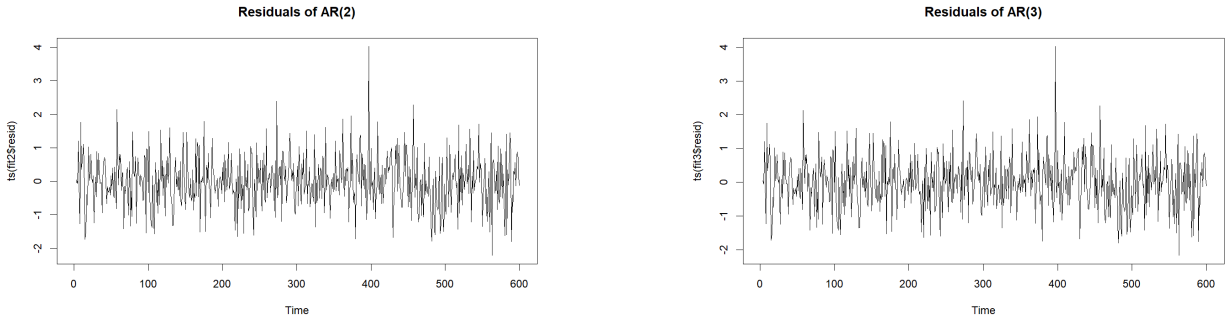
Let us rigorously test AR(2) and AR(3) to verify they behave like white noise. Recall our three-fold criterion for white noise identification: **(i)Uncorrelated sequence:** The ACF of the residuals for both p=2 and p=3 appear to be highly uncorrelated, satisfying this condition. **(ii)Zero mean:** This can be determined by a hypothesis test. Since the sample size is large ($n \approx 600$), a normal diagnostic convention Z–test is appropriate for testing $H_0 : \mu = 0$ for the residual means. Under the large–sample approximation, the 95% confidence bounds correspond to ± 1.96, and we use the statistic

$$Z = \frac{\bar{x}}{\sqrt{\hat{\sigma}^2}}.$$

$$\text{AR(1): } Z_1 = \frac{0.007657975}{\sqrt{0.5765795}} = 0.0101 \quad (|Z_1| < 1.96),$$

$$\text{AR(2): } Z_2 = \frac{0.007935085}{\sqrt{0.5769402}} = 0.0104 \quad (|Z_2| < 1.96).$$

We fail to reject H_0, and both residual means are consistent with white noise. *Note:* A classical test uses $Z = \bar{x}/(\hat{\sigma}/\sqrt{n})$, but the usual diagnostic form compares $\bar{x}$ to $\sqrt{\hat{\sigma}^2}$; with $n \approx 600$ the difference is negligible. **(iii)Constant variance:** The residuals of the autoregressions display homoscedasticity (Figure 5).



(a) Residuals AR(2) *(b) Residuals AR(3)*

Figure 5: Homoscedastic residuals of AR(2) and AR(3) models.

It appears both of the models are behaving like white noise. Although $\hat{\phi}_3$ of $AR(3)$ seem like it might be a negligible coefficient. There is potential overfitting present. To test whether the third coefficient in the AR(3) model is significant, we evaluate $H_0 : \phi_3 = 0$ using its asymptotic variance. From the output,

$$\hat{\phi}_3 = 0.0287, \qquad \widehat{\text{Var}}(\hat{\phi}_3) = 0.0016764682,$$

giving

$$Z = \frac{0.0287}{\sqrt{0.0016764682}} = 0.7015.$$

Since $|Z| < 1.96$, $\hat{\phi}_3$ is not significant, indicating that the AR(3) model overfits relative to AR(2).

Akaike information criterion

To verify our chosen AR(2) model and ensure parsimony, the AIC can give us an idea of the ideal order (p) that minimises information loss for our AR model.

$$\text{AIC values: } \; p = 0: \; 1047.509377, \quad p = 1: \; 38.868466, \quad p = 2: \; 0.000000, \quad p = 3: \; 1.504829.$$

AR(2) is the most appropriate model as it corresponds to the lowest AIC, representing the minimum estimated Kullback-Leibler (KL) divergence from the true, underlying data-generating process.

4 Final model

Final ARIMA model

Using `arima(Y, order = c(2,0,0))`—chosen because the AR(2) Residuals were closest to white noise—we re-fit the AR parameters for the stochastic component of the series. This yields the Y estimate AR(2) model

$$Z_t = c + \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma^2),$$

with parameter estimates (standard errors in parentheses):

$$c = -0.0335\,(0.4257), \qquad \phi_1 = 0.6643\,(0.0393), \qquad \phi_2 = 0.2649\,(0.0394), \qquad \sigma^2 = 0.576.$$

Since the intercept is not significant

$$t = \frac{-0.0335}{0.4257} \approx -0.08 < 1.96,$$

it is absorbed into the deterministic mean component, and the fitted stochastic model becomes

$$Z_t = 0.6643\, Z_{t-1} + 0.2649\, Z_{t-2} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, 0.576).$$

From (iii), the original series is decomposed as

$$X_t = \sum_{i=0}^{11} s_i\, \delta_{t,i} + Y_t, \qquad Y_t \approx Z_t,$$

and considering the estimated seasonal dummy coefficients from (v):

$$\begin{aligned} S_t = {} & 1.1217\,\text{jan}_t + 2.3694\,\text{feb}_t + 2.8675\,\text{mar}_t + 1.7260\,\text{apr}_t + 1.6127\,\text{may}_t + 1.3839\,\text{jun}_t \\ & + 0.3643\,\text{jul}_t + 0.4093\,\text{aug}_t + 3.3945\,\text{sep}_t + 4.8671\,\text{oct}_t + 4.9617\,\text{nov}_t + 5.1324\,\text{dec}_t. \end{aligned}$$

Final model for the original series X_t:

$$\boxed{X_t = S_t + 0.6643\, Z_{t-1} + 0.2649\, Z_{t-2} + N_t, \qquad N_t \sim \mathcal{N}(0, 0.576).}$$

Appendix: R Code

```
load("C:/Users/MB/Downloads/pond.RData")

# Verifying the data is for 12 months
head(X, n = 12)

# Basic plot but remove the default x-axis
plot(X, type = "l", main="Pond Water Levels")

## Fit a linear trend
XX = 1:length(X)
fit1 = lm(X ~ XX)

trend1 = fitted(fit1)  # Trend as a vector
trend1 = ts(trend1, start = start(X), frequency = frequency(X))

## Superimpose on time series plot
lines(trend1, col = "blue")

#########################################################################################################

## The regression appears almost completely flat! Let us verify whether the slope is tiny
coef(fit1)
summary(fit1)

# The P values show the intercept is highly significant but the slope isn't (if there is a trend we wan
# However, we can clearly observe the significant dip in the time series corresponding to the late 2000s
# It appears there might be seasonality semi-annually or annually.
# Plotting ACF to check
acf(X)
acf(ts(X))

# The ACF has a very high lag 1 value, meaning each observation strongly depends on the previous one.
# There also appear to be periodic spikes every 12 lags or so, I speculate potential annual seasonality
# The slow decay of the ACF, instead of an abrupt cutoff, indicates that the series has underlying persi
#But we wont know until testing signifiance of parameters
#########################################################################################################

## Finally, fit seasonal effect

## Create dummy variables for each month
n = length(X)
jan = as.numeric((1:n %% 12) == 1)
feb = as.numeric((1:n %% 12) == 2)
mar = as.numeric((1:n %% 12) == 3)
apr = as.numeric((1:n %% 12) == 4)
may = as.numeric((1:n %% 12) == 5)
jun = as.numeric((1:n %% 12) == 6)
jul = as.numeric((1:n %% 12) == 7)
aug = as.numeric((1:n %% 12) == 8)
sep = as.numeric((1:n %% 12) == 9)
oct = as.numeric((1:n %% 12) == 10)
nov = as.numeric((1:n %% 12) == 11)
dec = as.numeric((1:n %% 12) == 0)

## Fit model to our Time Series using the dummy variables
```

```
fit2 = lm(X ~ 0 + jan + feb + mar + apr + may + jun + jul + aug +
            sep + oct + nov + dec)

fit2#Yields us our coefficients


## Now see what variation is left after removal of seasonal effects.
seasonal = ts(fitted(fit2), start=start(X), end=end(X),
              frequency=frequency(X))

# Obtaining de-seasonalised, detrended "Y" residuals
Y = X - seasonal

plot(X, ylab="monthly level", main="Seasonal effect")
lines(seasonal, col="blue")
plot(Y, ylab="Residuals", main="De-seasonalised series (Y)")

summary(fit2) #checking the statistical validity of our method

# After removing linear trend and seasonal effects, the residual series Y shows no obvious remaining sea
# It contains short-term autocorrelation and a noticeable structural dip in the late 2000s,
# which represents an irregular shock (see how its outside the predicted seasonal trend) rather than a s

##############################################################################################################

acf(Y, main = "ACF of Y")
plot(Y, ylab = "Residuals")

# The residual series Y displays strong short-term autocorrelation but no remaining seasonal pattern or
# We again observe the prominent dip in the late 2000s { one-off, rather than structural.
# The overall variance appears roughly constant aside from this local disturbance.
# The ACF starts very high at lag 1 and then decays gradually and smoothly over many lags.
# That pattern is typical of an AR(p) process, rather than an MA process (often AR(1) or AR(2)):
# AR models have a slow, roughly geometric decay in the ACF, with a slowly decaying ACF characteristic.
# A pure MA(q) process would instead show a few significant spikes and then cut off sharply after lag q

##############################################################################################################

# To finalise the order p for our AR(p) process we can check the PACF of Y.
# The PACF cuts off after lag p, indicating the appropriate order.
# Based on the ACF/PACF behaviour and parsimony considerations,
# we can determine an appropriate p value for our model AR(p).
pacf(Y, main = "PACF of Y" )

# Seems like the PACF plot "cuts off" after lag 2!
# Thesis: This is textbook pattern for a low order p AR(p) model.
# However, a pacf alone is not enough. let us test this by comparing the AR(p) model for p = 1, 2, 3.

##############################################################################################################

# Method: Solving Yule-Walker Equations "by hand"
# Estimating AR(p) coefficients using Yule{Walker system
acf(Y, plot = FALSE)$acf[2:4]  # Computing sample autocorrelations (index 1 is lag 0)

# Solving Yule{Walker equations for AR(1)
rho = acf(Y, plot = FALSE)$acf
rho1 = rho[2]
```

```
alpha1_AR1 = rho1  # For AR(1) alpha just equals sample p
alpha1_AR1

# Solving Yule{Walker equations for AR(2)
rho2 = rho[3]
R2 = matrix(c(1, rho1, rho1, 1), nrow = 2, byrow = TRUE)  # Toeplitz matrix of order 2 for sample p
Rvec2 = c(rho1, rho2)

alpha_ar2 = solve(R2, Rvec2)
alpha_ar2

# Solving Yule{Walker equations for AR(3)
rho3 = rho[4]
R3 = matrix(c(1, rho1, rho2,
              rho1, 1, rho1,
              rho2, rho1, 1), nrow = 3, byrow = TRUE)  # Toeplitz matrix of order 3 for sample p
Rvec3 = c(rho1, rho2, rho3)

alpha_ar3 = solve(R3, Rvec3)
alpha_ar3

# The fitted AR(1) model is (insert LaTeX here) AR(2) model is AR(3) model is
fit1=ar(Y, order.max=1, aic=F,) #This will help us verify whether our matrix method was valid
fit1

fit2=ar(Y, order.max=2, aic=F,) #This will help us verify whether our matrix method was valid
fit2

fit3=ar(Y, order.max=3, aic=F,) #This will help us verify whether our matrix method was valid
fit3
##########################################################################################################
# The next step is to compute the corresponding AR(p) residuals.

# Plotting the AR(p) residuals and their ACF
acf(ts(fit1$resid), na.action = na.omit, main="ACF of residuals; AR(1)",) #Plotting the acf of residuals


acf(ts(fit2$resid), na.action = na.omit, main="ACF of residuals; AR(2)") #Plotting the acf of residuals


acf(ts(fit3$resid), na.action = na.omit, main="ACF of residuals; AR(3)") #Plotting the acf of residuals

mean1 = mean(fit2$resid[3:length(X)])
mean1
var1 = var(fit2$resid[3:length(X)])
var1
mean1/sqrt(var1)

mean2 = mean(fit3$resid[4:length(X)])
mean2
var2 = var(fit3$resid[4:length(X)])
var2
mean2/sqrt(var2) #testing for signifiance (white noise condition mean of residuals of ACF should be 0)
#Z-test since large sample, well within 1.96, so we accept the null mean = 0

fit3$asy.var.coef #Testing for insignificance of 3rd coeff
```

```
fit3$ar[3]/sqrt(fit3$asy.var.coef[3,3]) # Test statistic (less than 1.96)

#for constant variance it is suffience to examine plot of time series of the residuals
plot(ts(fit2$resid), main="Residuals of AR(2)")

plot(ts(fit3$resid), main="Residuals of AR(3)")
# ------------------------------------------------------------
# AR(1) residuals
# ------------------------------------------------------------
# - Residuals still show strong visible autocorrelation
#   (slow mean reversion, long drifts).
# - The ACF has many significant spikes across many lags.
# - Interpretation: AR(1) fails to remove dependence.
# - Residuals are not white noise → AR(1) is inadequate.

# ------------------------------------------------------------
# AR(2) residuals
# ------------------------------------------------------------
# Residual ACF is much smaller and mostly within the significance bounds.
# Residual time series looks closer to white noise with no obvious remaining structure.
# Interpretation: AR(2) removes almost all serial dependence.
# Residuals behave like white noise → strong candidate.

# ------------------------------------------------------------
# AR(3) residuals
# ------------------------------------------------------------
# Similar to AR(2): ACF shows only tiny correlations, all within significance limits.
# No remaining visible structure in residual plot.
# Interpretation: AR(3) fits, but adds little improvement.
# Third coefficient is very small (~0.0288) → negligible effect.
# AR(3) likely overfits and does not meaningfully improve on AR(2).

# The conclusion above is in line with our original thesis from our pacf() plot!
# AR(2) provides the best balance:
# - Residuals appear white noise.
# - No remaining serial correlation.
# - Avoids unnecessary parameters (unlike AR(3)).

# Therefore, AR(2) is selected as the best-fitting model.

#let us verify this with:
fit4=ar(Y, order.max=3, aic=T,) #Settign aicf=T returns the best fit out of up to order 3
fit4
fit4$aic #Returns the total list of aic up to 3

# We refer to the AR(2) residuals as Z.

##########################################################################################################
#Re-fitting chosen model using arima()

fit_ar2_final <- arima(Y, order = c(2, 0, 0))
fit_ar2_final
summary(fit_ar2_final)

# The ARIMA output reports each estimated parameter together with its
# standard error (s.e.), which describes the uncertainty of the estimate.
# A coefficient that is small relative to its s.e. is not statistically
```

```
# different from zero.

# In our fitted AR(2) model for Y_t, arima() returned:
#   ar1       = 0.6643   (s.e. = 0.0393)
#   ar2       = 0.2649   (s.e. = 0.0394)
#   intercept = -0.0335  (s.e. = 0.4257)

# The intercept is extremely small relative to its s.e.:
#   -0.0335 / 0.4257  -0.08,
# far below the |z| > 2 significance threshold. Thus, the intercept is
# not significantly different from zero.

# This is exactly what we expect: after detrending and deseasonalising,
# Y_t was constructed to have mean approximately zero. Therefore the true
# AR(2) model should contain no meaningful intercept. The ARIMA output
# confirms this by estimating an intercept essentially equal to zero.

# The AR(2) coefficients are both highly significant and closely match the
# Yule{Walker estimates, with very small standard errors. This supports
# the adequacy of the AR(2) structure for modelling the stochastic
# component of the series.

# Final stochastic model:
#   Y_t = 0.6643 * Z_{t-1} + 0.2649 * Z_{t-2} + N_t,
#   N_t ~ N(0, ^2),  ^2  0.576.


# S_t = The resulting estimated seasonal component (S_t) is:
#S_t = 1.1217 * jan_t + 2.3694 * feb_t + 2.8675 * mar_t + 1.7260 * apr_t +
#1.6127 * may_t + 1.3839 * jun_t + 0.3643 * jul_t + 0.4093 * aug_t +
#3.3945 * sep_t + 4.8671 * oct_t + 4.9617 * nov_t + 5.1324 * dec_t

#Final Model: X_t = S_t + 0.6643 * Z_{t-1} + 0.2649 * Z_{t-2} + N_t,
```